

Sustainability goals: IoT solutions help industries meet sustainability targets by optimising energy use and reducing carbon footprints.

- **Asset tracking**

Real-time location: Azure IoT enables real-time tracking of assets across facilities, improving asset utilisation and reducing losses.

Condition monitoring: Sensors can monitor the condition of assets, ensuring they are maintained properly and reducing the risk of damage.

Operational efficiency: Improved asset tracking leads to enhanced operational efficiency and reduced operational costs.

- **Quality control and assurance**

Automated inspections: IoT devices can automate the inspection process, ensuring consistent quality checks and reducing human error.

Data-driven insights: By analysing data from production lines, companies can identify defects early and take corrective actions promptly.

Continuous improvement: Ongoing data collection and analysis support continuous improvement initiatives, enhancing product quality over time.

- **Remote monitoring and management**

Facility management: Azure IoT allows for the remote monitoring of facilities, enabling quick responses to issues without the need for onsite personnel.

Operational visibility: Enhanced visibility into operations helps in identifying inefficiencies and optimising processes.

Scalability: IoT solutions can be scaled to include additional devices and sensors as business needs grow, providing flexibility and adaptability.

- **Enhanced safety and compliance**

Workplace safety: IoT devices can monitor environmental conditions and alert workers to potential hazards,

improving workplace safety.

Regulatory compliance: Automated data collection and reporting ensure compliance with industry regulations, reducing the risk of penalties.

Incident response: Real-time alerts and monitoring enable swift incident response, minimising the impact of safety breaches.

How to develop IoT solutions with Azure

Developing data-intensive IoT solutions with Microsoft Azure involves a strategic approach to harnessing the power of connected devices and advanced analytics. It starts by defining clear objectives and identifying key data sources, such as sensors and IoT devices, to ensure comprehensive data collection. You can utilise Azure IoT Hub to securely connect, manage, and ingest data from various endpoints.

- Implement robust data storage solutions like Azure Data Lake or Azure Blob Storage to handle large volumes of structured and unstructured data efficiently. Leverage Azure Stream Analytics for real-time data processing and Azure Machine Learning to derive predictive insights, enhancing decision-making capabilities.
- Ensure data security and compliance by integrating Azure Security Center and Azure Active Directory for identity management and threat protection. Optimise performance and scalability with Azure Functions and Azure Logic Apps, enabling seamless integration and automation across services.
- Finally, visualise insights using Power BI, providing stakeholders with interactive and actionable dashboards.

By strategically combining these Azure services, businesses can develop scalable data-driven IoT solutions that drive innovation and operational excellence.

Scalable event-driven architecture (EDA) using Azure Event Grid

Event-driven architecture (EDA) is an architectural pattern commonly used for event processing, enrichment, consumption and reaction to events. Asynchronous message processing can be achieved using EDA at scale. It can also help to broadcast events to multiple consumers (event handlers) at the same time to execute the triggered action.

In cloud platforms like Azure, enterprise application architecture requires scalable, high performing event driven engines like Azure Event Grid. The latter can simplify event delivery, build reliable cloud applications, and enable inter-service communication as a seamless experience.

In an Azure event-driven architecture, event producers can be an IoT hub, machine learning service, cloud event sources, Azure Service Bus, or Azure Blob Storage, to name a few. These events are broadcast to multiple consumers registered with an event grid. These event consumers can be an Azure cloud function, logic apps, event hubs, storage queues or application services like the Azure Automation service.

Each event handler that receives an event and processes it needs to subscribe to event topics. The topic is the endpoint where event sources send/publish the events. When the event grid receives the events, it finds the list of subscribed handlers to route events, and triggers the event handling action appropriately.

An event grid has advanced event filters which make sure that only relevant events are received by event handlers. Sending events to more than one event handler is called fan-out. An event grid makes multiple copies of events to as many handlers as needed.

Azure Event Grid is highly reliable since it ensures delivery of events even when there are high-